Beam Source Design: Growth Rate, Dopant Density & Microcapillary Array

Scope. This document follows the design math used in `Oven_Calculations_2_5.nb`. It sets the operating point $(T_{\text{oven}}, P_{\text{Ar}})$ and the microcapillary geometry (N, D, L) needed to hit a target Ar growth rate and Rb dopant density.

Design philosophy. Follow the Weinstein/Regmi procedure [1] for setting oven temperature and pressure from a target Rb:Ar ratio and crystal growth rate, but replace the single-orifice nozzle with a hexagonally packed microcapillary array in the style of Senaratne *et al.* [2] for atomic-beam collimation. Because Ar is the dominant species ($> 10^5\times$ the Rb number flux), the capillary geometry is sized to deliver the desired Ar throughput; the Rb flow is then a small, passive perturbation whose partial pressure follows from the Rb reservoir temperature.

Constants and parameters used throughout:

Quantity	Symbol	Value
Boltzmann	k_B	1.381×10^{-23} J/K
Avogadro	N_A	6.022×10^{23} /mol
Ideal gas constant (Torr)	R	62.364 L·Torr/(mol·K)
Ar molar mass	M_{Ar}	39.948 g/mol
Ar hard-sphere diameter	σ	3.35×10^{-10} m (mean-free-path), 3.35 Å (NIST viscosity)
Ar critical temperature	T_c	150.687 K
Ar critical density	ρ_c	13.407 mol/dm^3
Solid Ar density (20 K)	$\rho_{\text{s,Ar}}$	1.768 g/cm^3
Crystal target radius	r_c	$0.2\text{ in} = 0.508\text{ cm}$

1. Target Operating Point: Rb:Ar Ratio

Inside the oven the Rb partial pressure is set by the vapor-pressure curve of liquid Rb at the reservoir temperature, while P_{Ar} is set by the backfill line. The dopant ratio in the vapor is

$$\frac{n_{\mathrm{Rb}}}{n_{\mathrm{Ar}}} \;=\; \frac{P_{\mathrm{Rb}}(T_{\mathrm{oven}})}{P_{\mathrm{Ar}}}.$$

Because solidification in the thin-film regime is dominated by sticking (not diffusive reshuffle), the vapor ratio is the ratio that ends up trapped in the crystal, to within $\mathcal{O}(1)$ factors set by the relative sticking coefficients.

Rb vapor pressure (Antoine form for liquid Rb, Nesmeyanov):

$$P_{\mathrm{Rb}}(T) \;=\; 760 \cdot 10^{\,A + B/T} \quad [\mathrm{Torr}], \qquad A = 4.312,\; B = -4040.$$

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2. Dopant Density in the Solid

Number density of Ar atoms in the solid at 20 K:

$$n_{\mathrm{Ar}}^{\mathrm{solid}} \;=\; \frac{\rho_{\mathrm{s,Ar}}\, N_A}{M_{\mathrm{Ar}}} \;\approx\; 2.66 \times 10^{22}\ \mathrm{atoms/cm^3} \;=\; 2.66 \times 10^{19}\ \mathrm{atoms/mm^3}.$$

Assuming the Rb:Ar vapor ratio is preserved on deposition, the Rb number density in the solid is

$$n_{\mathrm{Rb}}^{\mathrm{solid}} \;=\; \frac{P_{\mathrm{Rb}}(T_{\mathrm{oven}})}{P_{\mathrm{Ar}}} \cdot n_{\mathrm{Ar}}^{\mathrm{solid}}.$$

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3. Microcapillary Array: Geometry & Count

The capillaries are hexagonally packed in an equilateral-triangular boundary (Senaratne-style [2]): the boundary forces hex registration and the triangular footprint makes the capillary count a triangular number. Microcapillaries are cut from 33RW gauge 304-SS hypodermic tubing (OD $d_o = 0.008$ in, ID $D = 0.004$ in $= 100\ \mu$m).

Triangle side length $S = \sqrt{3} \cdot 0.1$ in ≈ 0.1732 in (inscribed circle of radius 0.05 in). Rows and total count:

$$n_r \;=\; \left\lfloor \frac{S}{d_o} \right\rfloor \;=\; 21, \qquad N \;=\; \frac{n_r(n_r+1)}{2} \;=\; 231.$$

Capillary length $L = 0.300$ in $= 0.762$ cm (aspect ratio $L/D \approx 76$).

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4. Argon Viscosity (NIST Correlation)

For the Hagen-Poiseuille conductance below, the viscosity of Ar must be evaluated at the oven conditions. NIST RefProp uses a dilute-gas term plus a density-dependent residual (Lemmon & Jacobsen):

Collision integral:

$$\Omega(T) \;=\; \exp\left[\sum_{i=0}^{4} b_i \left(\ln \tfrac{T}{r}\right)^i\right], \quad r = 143.2\ \mathrm{K}.$$

Coefficients: $b_0 = 0.431,\ b_1 = -0.4632,\ b_2 = 0.08406,\ b_3 = 5.341\times10^{-3},\ b_4 = -3.31\times10^{-3}$.

Dilute-gas viscosity (μPa·s):

$$\eta_0(T) \;=\; \frac{0.026696\,\sqrt{M_{\mathrm{Ar}}\,T}}{\sigma^2\,\Omega(T)}.$$

Residual viscosity term (dimensionless reduced $\tau = T_c/T, \delta = \rho/\rho_c$):

$$\Gamma(T,\rho) \;=\; \sum_{i=1}^{6} N_i\,\tau^{t_i}\,\delta^{d_i}\,\exp\!\left[-\gamma_i\,\delta^{l_i}\right].$$

Total dynamic viscosity: $\eta(T,\rho) = \eta_0(T) + \Gamma(T,\rho)$.

Ideal-gas density needed for δ:

$$\rho(T,P) \;=\; \frac{P\cdot 133.322}{R_{\mathrm{SI}}\,T\cdot 1000}\quad [\mathrm{mol/dm^3}], \quad R_{\mathrm{SI}} = 8.314\ \mathrm{J/(mol\backslash cdotpK)}.$$

(Numerically $\eta \approx 22.5\ \mu$Pa·s at 300 K, 10 Torr; the residual Γ is $< 1\%$ at design pressures.)

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5. Flow Regime: Mean Free Path & Knudsen Number

$$\lambda(T,P) \;=\; \frac{k_B T}{\sqrt{2}\,\pi\,\sigma^2\,P}.$$

Knudsen numbers set the regime:

$$\mathrm{Kn}_D \;=\; \lambda / D, \qquad \mathrm{Kn}_L \;=\; \lambda / L,$$

with $\mathrm{Kn} < 0.01$ viscous, $0.01 < \mathrm{Kn} < 1$ transitional, $\mathrm{Kn} > 1$ molecular. In the design regime below the capillary is molecular at the inlet ($\mathrm{Kn}_D \gg 1$) and transitional over its length ($\mathrm{Kn}_L \lesssim 1$).

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6. Viscous Conductance (Hagen-Poiseuille Limit)

For a long, cylindrical tube with $\mathrm{Kn} \ll 1$:

$$C_{\mathrm{visc}} \;=\; \frac{\pi\, D^4\, P_{\mathrm{avg}}}{128\, \eta\, L}.$$

With D, L in cm, P_{avg} in Torr, η in μPa·s, this reduces to the compact form used in the notebook:

$$C_{\mathrm{visc}}[\mathrm{L/s}] \;=\; \frac{32600}{10\,\eta(T,\rho)} \cdot \frac{D^4}{L}\, P_{\mathrm{avg}}.$$

For an array of N parallel capillaries, $C^{(N)}_{\mathrm{visc}} = N \cdot C_{\mathrm{visc}}$. The inlet-to-outlet average $P_{\mathrm{avg}} \approx P_{\mathrm{oven}}/2$ is used (outlet opens into vacuum).

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7. Molecular Conductance (Knudsen Long-Tube Limit)

For $\mathrm{Kn} \gtrsim 1$ the Clausing/Knudsen long-tube result is

$$C_{\mathrm{mol}} \;=\; \frac{\pi}{12}\, \bar{v}\, \frac{D^3}{L}, \qquad \bar{v}(T) \;=\; \sqrt{\frac{8\, k_B\, T}{\pi\, m_{\mathrm{Ar}}}}.$$

In the notebook's working units (D, L in cm, $\bar{v}$ in cm/s, output in L/s) this is the familiar coefficient

$$C_{\mathrm{mol}}[\mathrm{L/s}] \;=\; 2.6 \times 10^{-4}\, \bar{v}\, \frac{D^3}{L} \quad \text{(per capillary)}; \qquad C^{(N)}_{\mathrm{mol}} \;=\; N \cdot C_{\mathrm{mol}}.$$

($2.6 \times 10^{-4} \approx \pi/12{,}000$, accounting for the $\mathrm{cm}^3 \to \mathrm{L}$ conversion.)

The conductance used for downstream flux calculations is the one appropriate to the regime. At $(T_{\text{oven}}, P_{\text{Ar}}) = (303\ \text{K}, 0.05\ \text{Torr})$ the molecular expression dominates by $\sim 300\times$, so $C \equiv C_{\text{mol}}^{(N)}$.

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8. Throughput, Atom Flux, and Crystal Growth Rate

Throughput (leaving the oven through the array):

$$Q \;=\; P_{\text{oven}}\, C \quad [\text{Torr}\cdot\text{L/s}].$$

Atom flux through the array:

$$\Phi \;=\; \frac{Q\, N_A}{R\, T_{\text{oven}}} \quad [\text{atoms/s}].$$

Mass flux and thickness growth rate on the cryogenic substrate (effective area πr_c^2, condensed to solid density $\rho_{\text{s,Ar}}$, assuming all atoms stick inside the capture solid angle):

$$\dot{m} \;=\; \Phi\, \frac{M_{\text{Ar}}}{N_A} \;=\; \frac{Q\, M_{\text{Ar}}}{R\, T_{\text{oven}}} \quad [\text{g/s}],$$

$$\dot{h} \;=\; \frac{\dot{m}}{\rho_{\text{s,Ar}}\, \pi\, r_c^2} \quad [\text{cm/s}].$$

In reporting units the notebook uses:

$$\dot{h}\,[\mu\text{m/hr}] \;=\; \frac{\dot{m}\,[\text{g/hr}]}{\rho_{\text{s,Ar}}\, \pi\, r_c^2} \cdot 10^4.$$

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9. Design Operating Point (Final Values)

Parameters:

T_{oven}	303 K
P_{Ar} (oven)	0.05 Torr $=$ 50 mTorr
N capillaries	231 (21 rows, triangular)
D, L	100 μm, 7.62 mm
L/D	76.2
r_c (crystal)	0.508 cm

Derived quantities:

λ_{Ar}	1.22 mm
Kn_D, Kn_L	12.2, 0.16 — **molecular at inlet, transitional over length**
$\eta_{\text{Ar}}(303\ \text{K})$	$\approx$ 22.8 μPa·s
$C^{(N)}_{\text{visc}}$	1.07×10^{-5} L/s (not applicable)
$C^{(N)}_{\text{mol}}$	3.13×10^{-3} L/s
Q_{Ar}	1.57×10^{-4} Torr·L/s
Φ_{Ar}	$\mathbf{5.0 \times 10^{15}}$ atoms/s
$\dot{h}_{\text{Ar}}$	**8.3** μm/hr
$n_{\text{Rb}}/n_{\text{Ar}}$	1.45×10^{-5}
$n^{\text{solid}}_{\text{Rb}}$	$\mathbf{3.86 \times 10^{17}}$ Rb/cm^3

The 5×10^{15} atoms/s Ar flux is consistent with the flow-rate anchor quoted in the lab notes.

Flags & Missing Pieces

1. Regime inconsistency between the two printed growth rates. The notebook prints both `growthRateGpH` (viscous) and `molecularGrowthRateGpH` (molecular). At the final design point, $\text{Kn}_D \approx 12$, so the viscous line (0.028 μm/hr) is not physically relevant — it is the molecular

value (8.3 μm/hr) that sets the design. I'd recommend: either gate the print by a Kn test, or clearly mark the viscous block as "legacy / single-orifice comparison only" and drop it from the final summary.

2. Capillary length inconsistent across notebook sections. The "Conductance Nozzle" block uses $L = 1$ cm, "All Together" uses 0.768 cm, and the 2-D contour plot section uses 0.762 cm. Since 0.762 cm $= 0.300$ in exactly, that is almost certainly the as-built value; the 0.768 is probably a typo. Reconcile to one value throughout (I've used 0.762 cm here).

3. Capillary ID inconsistent across notebook sections.0.01 cm in most blocks, 0.01016 cm ($= 0.004$ in, the 33RW nominal ID) in the contour-plot block. The latter is the spec value; the 0.6% difference propagates to $\sim 2.4\%$ in $C_{\mathrm{mol}} \propto D^3$ and $\sim 3.2\%$ in $C_{\mathrm{visc}} \propto D^4$.

4. Rb vapor-pressure curve evaluated below the melting point. The Antoine-form coefficients $(A, B) = (4.312,\ -4040)$ are the liquid-Rb coefficients (Nesmeyanov). Rb melts at 312.46 K; the design point is 303 K, i.e. ~ 10 K into the solid regime. For solid Rb near the melting point, Alcock/Honig give slightly different coefficients (e.g.\ $\log_{10}(P_{\mathrm{torr}}) = 4.857 - 4215/T$). At 303 K, the liquid formula overestimates P_{Rb} by $\sim 2\times$ relative to the solid curve — the reported $n_{\mathrm{Rb}}^{\mathrm{solid}}$ is probably high by that factor. Either (i) switch to the solid-Rb formula for $T < 312.46$ K, or (ii) run the Rb reservoir above the melting point (≥ 313 K) so the liquid formula is valid in its own domain.

5. Label error: ρ_c comment says "MPa". The NIST Ar critical density is 13.407 mol/dm^3, not MPa. Cosmetic, but worth fixing since it reads confusingly.

6. Missing: geometric collimation / divergence estimate. The capillary aspect ratio sets a geometric divergence half-angle $\theta_{1/2} = \arctan(D/L) \approx 0.75^\circ$ in the molecular-flow limit. Worth including alongside the beam-profile measurements (Method B gave $0.48 \pm 0.10^\circ$).

7. Missing: explicit Knudsen-number print. Notebook prints λ only, not Kn_D or Kn_L. One extra `Print` line per block would make the regime call self-documenting.

8. Missing: pressure-drop / P_{avg} check for the viscous branch. The viscous formula uses $P_{\mathrm{avg}} = P_{\mathrm{oven}}/2$ assuming $P_{\mathrm{outlet}} \to 0$. This is only true for continuum flow; for the molecular branch P_{avg} is irrelevant. Fine at the final operating point, but if the viscous output is kept for a higher-pressure comparison (e.g., the Weinstein $P_{\mathrm{Ar}} = 10$ Torr original design), the choice of P_{avg} should be stated explicitly.

9. Missing: capture-solid-angle / sticking efficiency in the growth rate. The derivation of $\dot{h}$ above assumes the entire throughput Q lands inside πr_c^2 with unit sticking. The geometric fraction depends on the source-to-substrate distance and the capillary beam divergence; a Senaratne-style multichannel source concentrates the flux into $\sim (D/L)^2$ sr, so the "all-atoms-captured" assumption is a closer fit here than for a 2π-sr single orifice, but it should still be

explicitly stated and bounded. This directly addresses the flow-rate vs. pressure-based flux tension noted in your lab record.

10. Missing: crystal radius rationale. $r_c = 0.2$ in is used without derivation; should be cross-referenced to the actual beam aperture / sapphire window geometry in Daniel's chamber.

References

[1] S. K. Regmi, *Optical Pumping of Rubidium Spin in a Solid Argon Matrix*, M.S. thesis, Univ. of Nevada, Reno, 2013 (Weinstein group). Oven / single-orifice nozzle design in §2.1.2.

[2] R. Senaratne *et al.*, "Effusive atomic oven nozzle design using an aligned microcapillary array," *Rev. Sci. Instrum.* **86**, 023105 (2015).

[3] NIST RefProp Ar viscosity correlation, https://trc.nist.gov/refprop/FAQ/NAO.PDF.